Answer all questions in the spaces provided

1(a) When an object is placed in water, it experiences an upthrust.

(i) Explain what is meant by upthrust. [1]

(ii) A container contains two liquids A and B which do not mix. The densities of liquids A and B, denoted by ρ_A and ρ_B, are 1.30×10^3 kg m^{-3} and 0.900×10^3 kg m^{-3} respectively. A cubical solid of side 0.200 m and mass 9.20 kg stays upright in equilibrium in between the two liquids as shown in Fig. 1.1. The depth of the solid submerged in Liquid A is denoted by *x*.

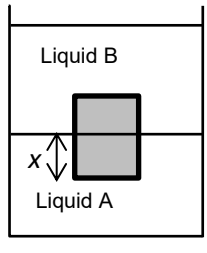

Fig. 1.1

1. Write down an expression for the upthrust exerted on solid by Liquid A in terms of *x*, ρ_A and *g*. [1]

2. Write down an expression for the upthrust exerted on solid by Liquid B in terms of *x*, ρ_B and *g*. [1]

3. Hence, or otherwise, determine *x*. [2]

(b) A hungry bear walks out from a cave onto a horizontal uniform beam of length L, supported on the left by a hinge attached to a wall, in an attempt to retrieve some "goodies" hanging at the right end (Fig. 1.2). The total weight of the bear and the beam is 900 N. The goodies weigh 80.0 N. Fig. 1.3 gives the tension T in the cable as a function of the bear's position given as a fraction x/L of the beam length.

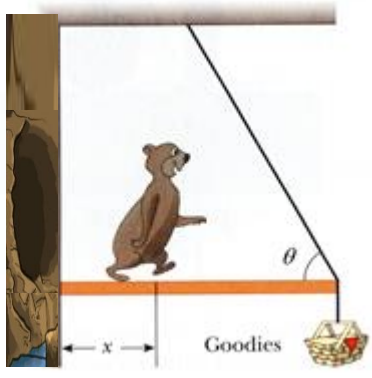

Fig. 1.2

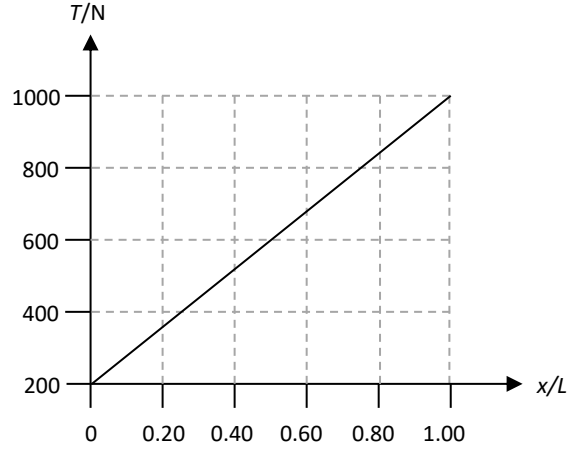

Fig. 1.3

(i) By considering the bear standing at the centre of the beam, calculate the angle θ.[2]

(ii) Suggest where, on the beam, the bear should stand so that his weight does not produce a moment on the beam. [1]

(iii) Calculate the mass of the beam. [2]